

Vineyard AI Field Challenge

Participant workflow

LEAD PARTNER MARCAJ



Concept illustration

01

Prepare imagery

02

AI pre-annotate

03

Review in Marcaj

04

Measure

05

Plan a route

06

Build the web app

Prepare the aerial imagery.

Split the image into tiles. Preserve coordinates and assign tile IDs.



Input: unannotated RGB orthomosaic in GeoTIFF format.



Output: georeferenced tiles with unique IDs.

Create AI pre-annotations.

Import polygons, bounding boxes and polylines into Marcaj.



1 **Canopies = polygons**

Class: vineyard. One polygon per distinguishable crown; exclude soil.

2 **Rows = polylines**

row_id: vine-row axis. interrow_id: passage axis. Link both to vineyard_id.

Waste = bounding box



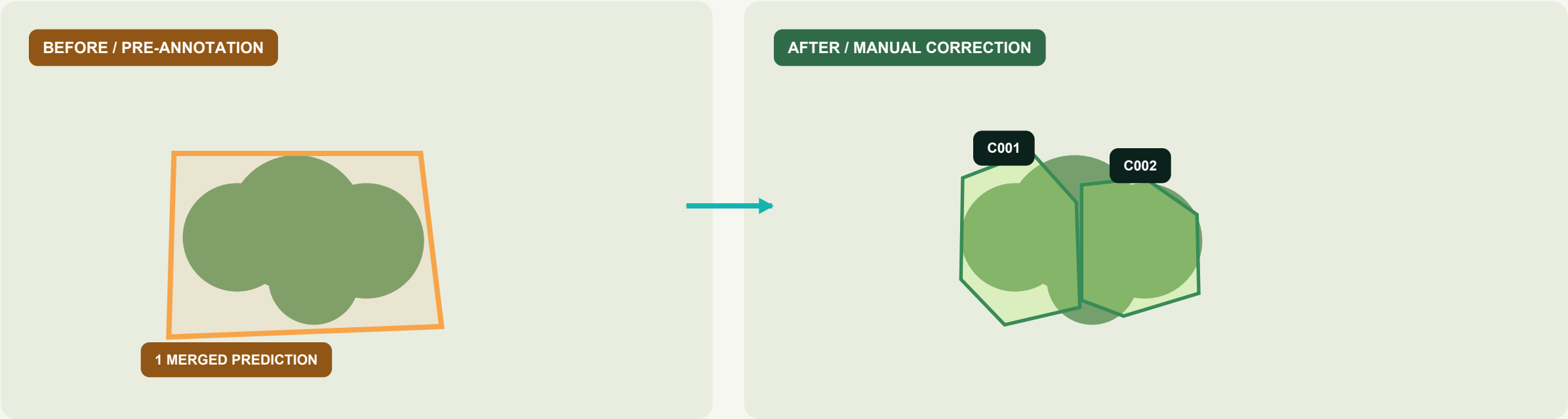
W01

Illustrative object.

Attributes: row structure = regular / disrupted / cannot assess; inter-row cover = bare soil / vegetation / mixed. Flag unreadable areas for exclusion.

Correct annotations in Marcaj.

Validate geometry and attributes. Export the corrected annotations.



Merged crowns and included soil.

Separate crowns, refine boundaries and retain IDs.

- Geometry**

Fix masks, boxes and row axes.
- Attributes**

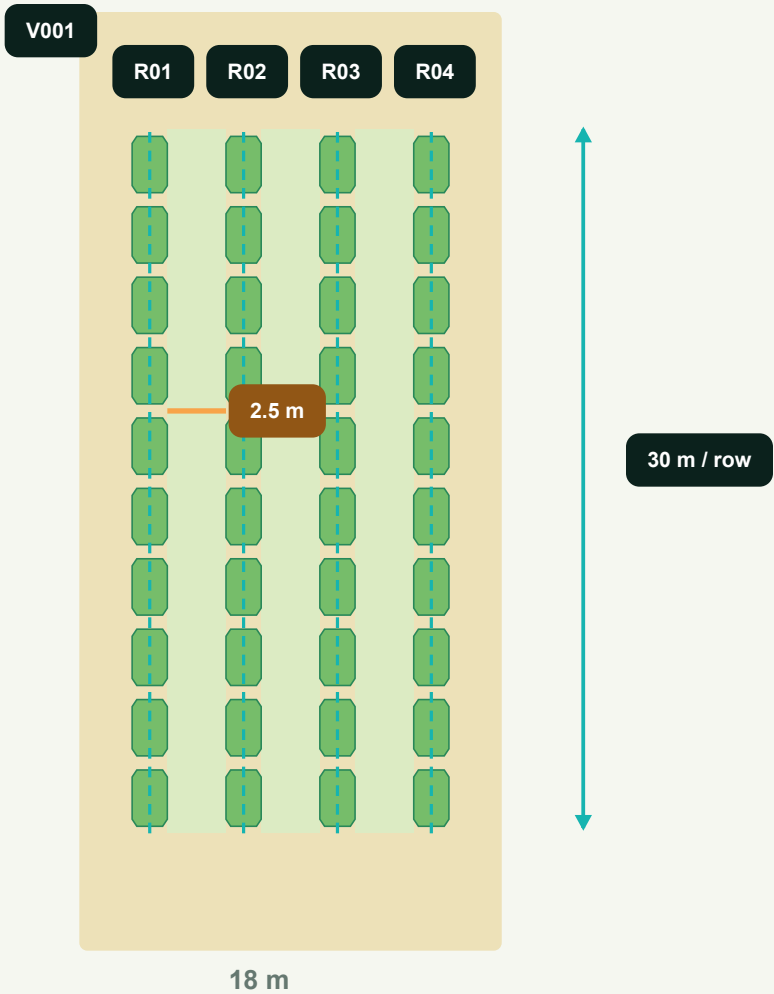
Check cover and row structure.
- Consistency**

Resolve duplicates at tile edges.
- Export**

Save corrected labels from Marcaj.

Calculate counts, lengths and areas.

Worked example in metres.



1
vineyard block: V001

120 m
total row length: 4 x 30 m

225 m²
inter-row area: 0.0225 ha

4 / 40
rows / canopy polygons

139.2 m²
canopy area: 0.01392 ha

720 m²
block extent: 18 x 40 m

Calculation rules
Canopy area: polygon union. Inter-row area: passage polygons.
Row length: axis polylines. Use polygons, not bounding boxes, for areas.

Map inspection stops and the route.

Set the start, target stops and return path.



~247 m

out-and-back route length

~1,095 m²

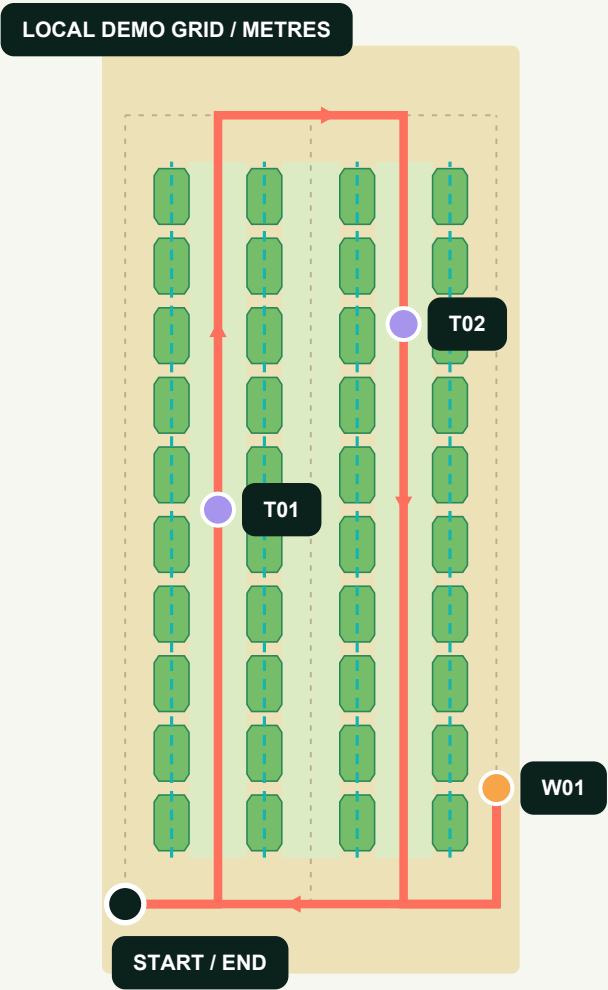
A001 analysis area

Route
START - T02 - T01 - START

Check
Passable paths, obstacles and target coordinates.

Minimize walking distance.

Visit all reachable targets and return to the organizer-defined start.



Route comparison

158 m

Separate return trips

110 m

Shortest closed tour



48 m less walking

30.4% reduction

● 3 / 3 targets

T01 + T02: inspect. W01: collect waste.

● Constraints

Avoid crowns and forbidden areas.

● Output

Route polyline, length and visited target IDs.

Build the web application.

Display objects, IDs, measurements and the route.

Vineyard AI / Field planner

ILLUSTRATIVE INTERFACE

Selected block

V001

Layers

- Canopy polygons
- Row axes
- Inspection targets
- Walking route

START / END

T01

T02

W01

Start: (1, 3) in demo metres

Route returns to its start.

110 m

route length

3 / 3

targets visited

139.2 m²

canopy area

225 m²

inter-row area

ROW ID	LENGTH	BLOCK
R01	30.0 m	V001
R02	30.0 m	V001
R03	30.0 m	V001
R04	30.0 m	V001

View annotation export

View route geometry

Submit the solution.

Required deliverables and scoring weights.

01

Model
Weights or a reproducible way to obtain them.

02

Marcaj annotations
Final export of corrected geometry and attributes.

03

Spatial data
Georeferenced polygons, axes, targets and route.

04

Measurements
Counts, row lengths and areas linked by ID.

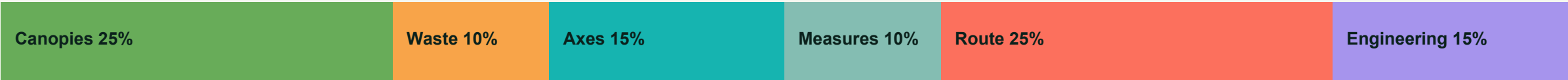
05

Web application
A working map, route and measurement views.

06

Reproducibility
Application code and processing instructions.

Scoring weights



85% objective output metrics

15% expert engineering assessment

Engineering criteria: architecture, robustness, scalability and performance.